Statistical Signal Processing
Lecture 04:
Optimal filtering and deconvolution (Wiener filter)

Literature and Problems: See course schedule

Agenda (preliminary):

Problem setup

- **time-domain signal estimation**
- **general model: blurring filter, noisy observations, Wiener filter**
- **demos 1D/2D signals**
- **modes: filtering, smoothing, prediction, deconvolution**
- **assumptions**
- **problem: minimize MSE = E [$(Y(n) - Y_{hat}(n))^2$]**

Discrete-time FIR Wiener filter

- **model**
- **FIR filter equation**
- **causal versus non-causal FIR filter**
- **minimize MSE → Wiener-Hopf equation**
- **performance: MSE_{min}**
- **WH-equation for filtering**
- **WH-equation for noiseless linear prediction**
- **WH-equation for linear prediction of noisy signal**

Discrete-time IIR Wiener filter (non-causal case)

- **model**
- **IIR filter equation**
- **minimize MSE → H(f) for Wiener filter**
- **special case: filtering, signal and noise uncorrelated**
- **interpretation of H(f) equation in terms of SNR**
- **performance: MSE_{min} calculated in time-domain and frequency-domain**

Deconvolution with discrete-time non-causal IIR-filter

- **model**
- **same task as before => same form of H(f)**
- **$S_{xx}(f)$, $R_{xy}(k)$ → $S_{xy}(f)$ gives H(f)**
- **interpretation of H(f) in terms of combined inverse filtering and frequency-domain weighted smoothing**